

Udacity

Marketing Analytics

Nanodegree Program
Project: Craft a Report

Objective Results Overview

Objectives

- Increase total sales by 30% on Black Friday 2018 vs 2017
- Decrease total ad spend by 30% from Black Friday 2017 to 2018

Objective 1 - Increase Total Sales by 30%

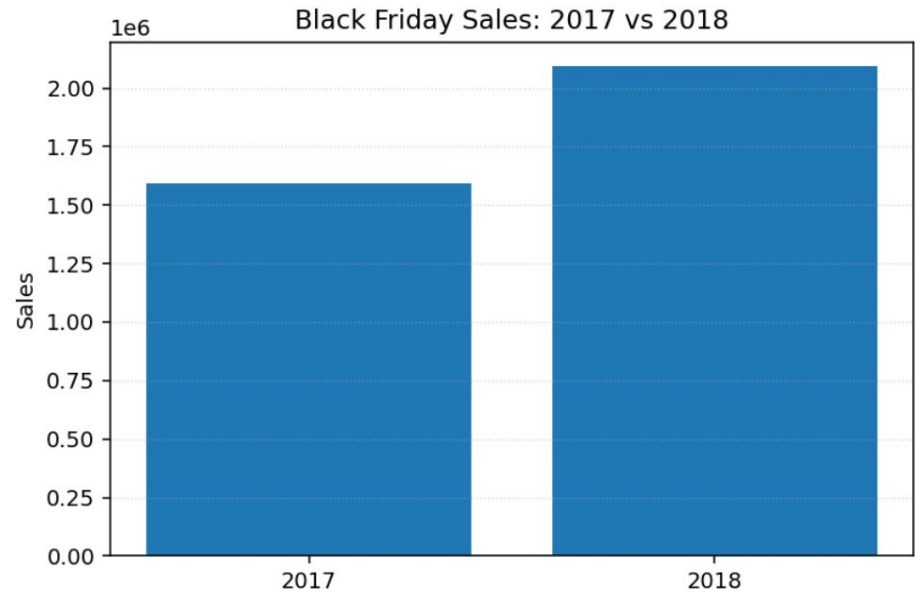
Objective: Increase total sales by 30% on Black Friday 2018 vs 2017.

Results:

- Total sales 2017: £1,594,913.85
- Total sales 2018: £2,092,431.46
- Year-over-year change: +31.2%

Conclusion: Objective met.

Black Friday Sales: 2017 vs 2018 Bar Chart



Why this visual: A bar chart clearly shows the change in sales between two time periods.

Objective 2 - Decrease Acquisition Cost by 30%

Objective: Decrease ad spend by 30% from Black Friday 2017 to 2018.

Note: Direct ad spend was not available, so Average CPA is used as a proxy for acquisition cost.

Results:

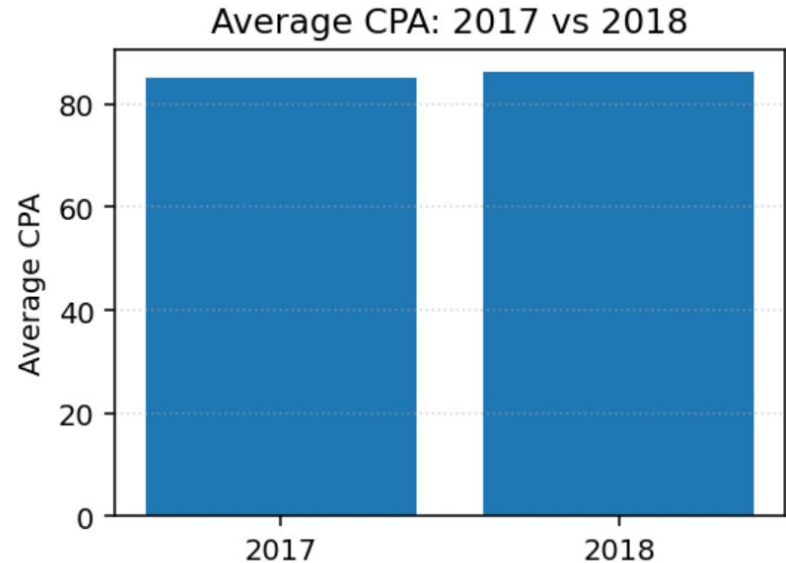
Average CPA 2017: £84.98

Average CPA 2018: £86.31

Change: +1.6%

Conclusion: Objective not met.

Average CPA: 2017 vs 2018 Bar Chart



Why this visual: A bar chart clearly shows the change in average CPA between the two time periods.

Objective Results - Summary

Objective Results Summary Table

Objective	Target	Result	Met?
Increase Sales	$\geq +30\%$	+31.2%	Yes
Decrease Spend	$\leq -30\%$	+1.6%	No

- Sales performance was strong and exceeded the growth target.
- Cost efficiency slightly worsened, indicating a need to improve acquisition efficiency.

Why this visual: A table clearly presents the values and allows direct comparison.

Evaluate the Audience

Research Question: How did sales vary by age range?

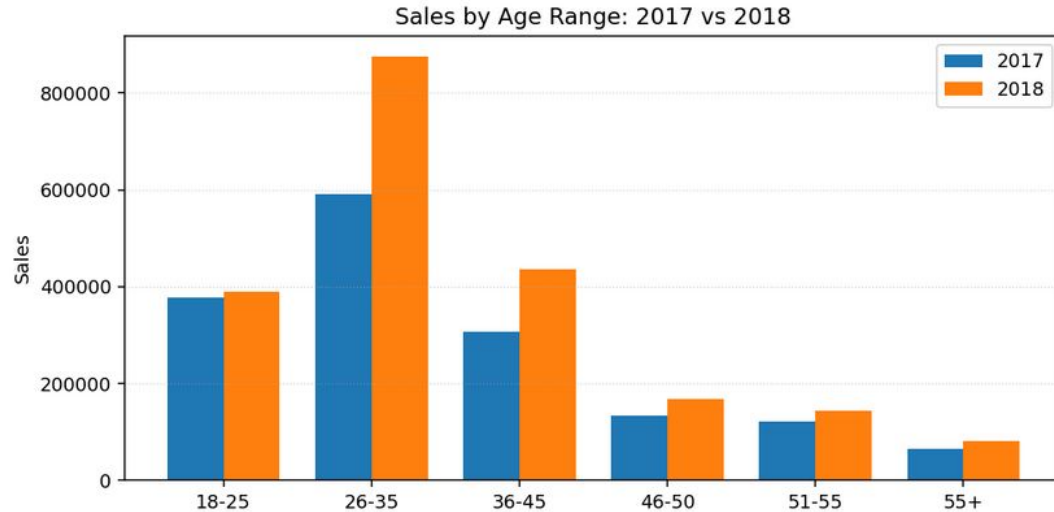
Finding

- The 26–35 age group generated the highest sales in both years.
- This segment also showed the strongest growth from 2017 to 2018.

Interpretation

- The 26–35 segment generates the most revenue and shows the strongest growth, making it the most valuable audience segment.

Sales by Age Range chart



Why this visual: A grouped bar chart makes it easy to compare sales across age ranges and between years.

Evaluate the Audience

Research Question: Which city generated the most sales?

Finding

- Houston was the top-performing city in both 2017 and 2018.
- Santa Barbara and San Antonio were also strong contributors.

Interpretation

- Marketing and inventory planning could prioritize Houston and similar high-performing markets.

Why this visual: A table is used here because it allows precise comparison of city-level sales values.

City sales table

	City	Year	total_sales
11	Houston	2018	277491.13
39	Santa Barbara	2018	257221.61
33	San Antonio	2018	240538.27
10	Houston	2017	207729.61
38	Santa Barbara	2017	199027.58
23	New York	2018	180799.95
17	Miami	2018	166885.68
32	San Antonio	2017	158325.67
16	Miami	2017	122034.88
22	New York	2017	120688.21

Evaluate the Marketing

Research Question: Was the ROI on our paid channel positive or negative?

Method

- Estimated paid spend = CPA × number of paid orders
- $\text{ROI} = (\text{Revenue} - \text{Estimated Spend}) / \text{Estimated Spend}$

Results

- Paid ROI 2017: +8.0%
- Paid ROI 2018: +6.7%

Conclusion

- Paid marketing was profitable in both years, although efficiency slightly declined in 2018.

Paid ROI summary table

Year	ROI (%)
2017	8.0
2018	6.7

Why this visual: A table clearly presents ROI values for each year and allows direct comparison.

Evaluate the Marketing

Research Question: Which age range had the best CPA?

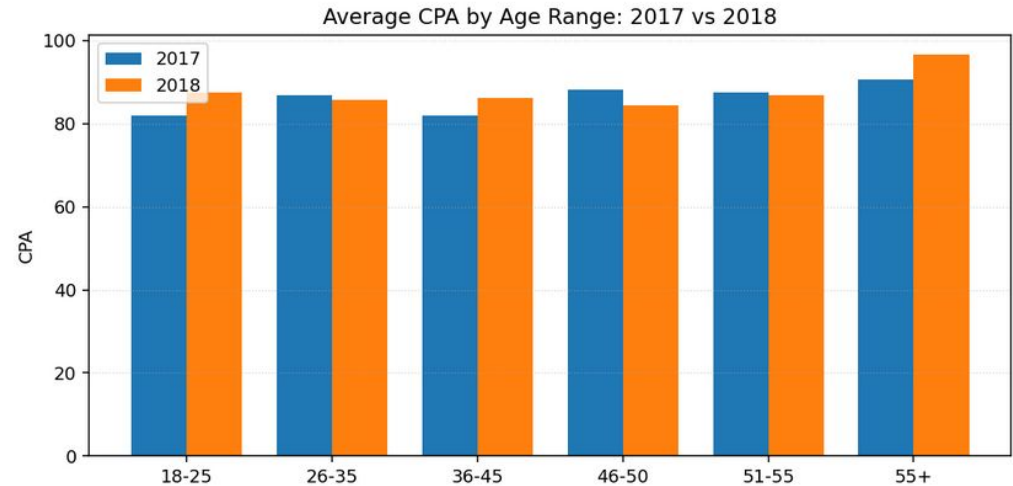
Finding

- The 18–25 and 36–45 segments had the lowest CPAs.
- The 55+ segment had the highest CPA.

Interpretation

- Younger and middle-aged customers are more cost-efficient to acquire.

Average CPA by Age Range chart



Why this visual: A bar chart highlights differences in average CPA across age groups.

Evaluate the Marketing

Research Question: Which channel drove the most sales?

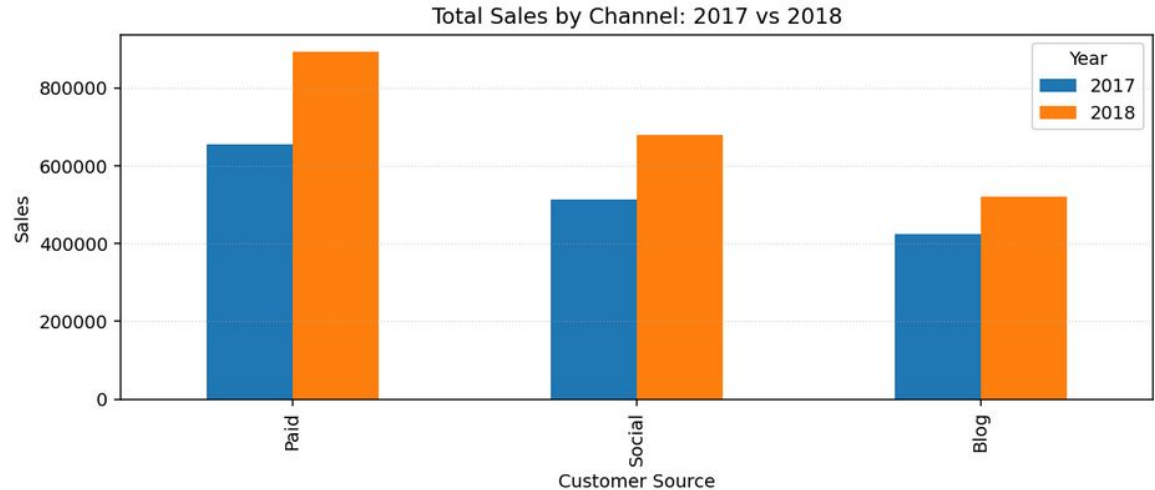
Finding

- Paid was the largest driver of sales in both years.
- Social grew significantly from 2017 to 2018.
- Blog content contributed steadily.

Interpretation

- Paid channels appear effective for volume, while social and content help support growth and engagement.

Total Sales by Channel chart



Why this visual: A bar chart is appropriate for comparing total sales across discrete marketing channels.

Evaluate the Sales

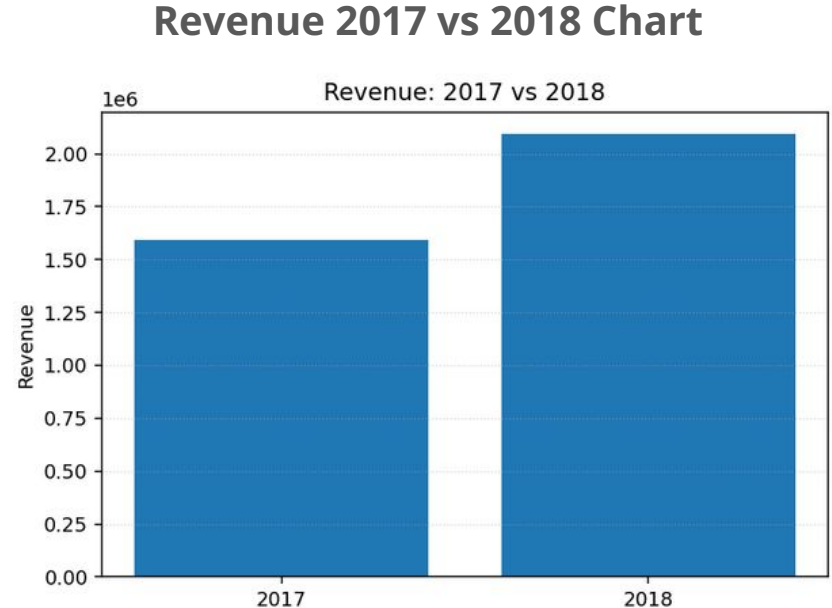
Research Question: How much revenue did we generate?

Results

- Revenue 2017: **£1.59M**
- Revenue 2018: **£2.09M**

Conclusion

- Revenue increased substantially, confirming overall campaign success.



Why this visual: A bar chart clearly shows the change in revenue between two time periods.

Evaluate the Sales

Research Question: Which customer spent the most?

Finding

- The highest-spending customer spent approximately £8,179 in 2018.

Interpretation

- A small number of high-value customers contribute disproportionately to revenue.

Why this visual: A table makes it easy to identify and compare individual high-spending customers.

Top Customer Table

	User ID	Year	total_sales
3775	1004277	2018	8179.00
1491	1001680	2017	8043.46
50	1000053	2017	7482.24
3924	1004448	2018	6127.63
2558	1002909	2018	6029.95
793	1000889	2017	6029.21
3112	1003539	2018	5148.08
5144	1005831	2018	4997.16
1719	1001941	2017	4965.65
1052	1001181	2017	4890.31

Evaluate the Product Categories

Research Question: Which product category was most popular?

Finding

- 2017: Pets was the top category.
- 2018: Grocery became the top category.

Interpretation

- Customer demand shifted slightly toward everyday essentials in 2018.

Why this visual: A table allows direct comparison of category-level sales values across years.

Tables of the Most Popular Category in 2017 and 2018

	Product	Category	Year	total_sales
8		Pets	2017	408474.65
6		Grocery	2017	398167.37
2		Clothing	2017	199092.78
0		Books	2017	198067.52
4		Games	2017	197660.34

	Product	Category	Year	total_sales
7		Grocery	2018	525438.17
9		Pets	2018	511712.65
3		Clothing	2018	267565.36
11		Toys	2018	267045.49
5		Games	2018	261043.60

Evaluate the Product Categories

Research Question: How did sales vary by category?

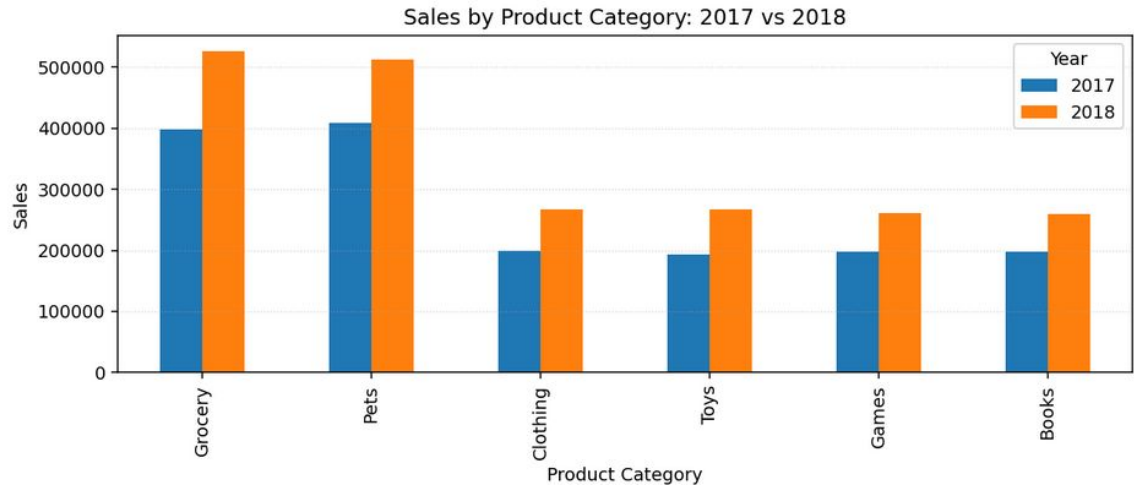
Finding

- Grocery, Pets, Clothing, and Toys all saw growth.
- Books remained stable.

Interpretation

- The portfolio is well balanced, with both essentials and discretionary items performing well.

Sales by Product Category Chart



Why this visual: A grouped bar chart allows comparison of category performance between years.

Everything Else

Data limitations

- No direct advertising spend data was provided in the dataset.
- Ad spend was therefore estimated using Average CPA as a proxy for acquisition cost.

Assumptions

- Each row represents a single order.
- CPA is treated as the average acquisition cost per order.
- Estimated spend = Average CPA $\times$ number of orders.

Notes

- Results represent Black Friday performance only and may not reflect full-year behavior.